

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

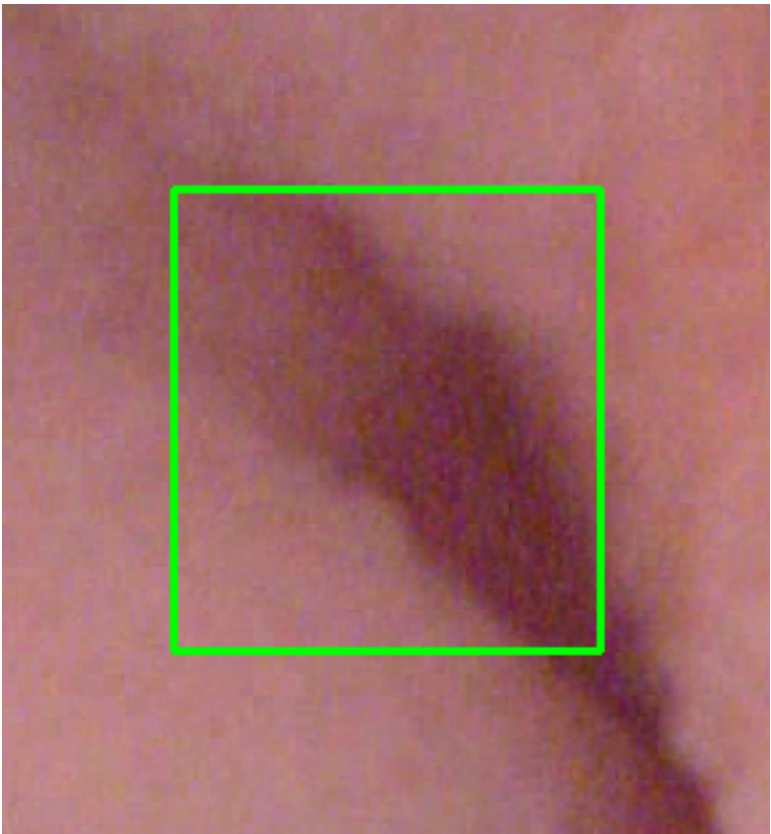
Video ID:	cd31e35a-59fd-4399-9f19-51eb485025d8
Total Frames:	3607
Frame Rate:	25.0 fps (assumed)
Duration:	144.3 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	54	✓
RT-DETR Detections	1278	✓
MedSAM2 Detections	1194	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 143 frames)



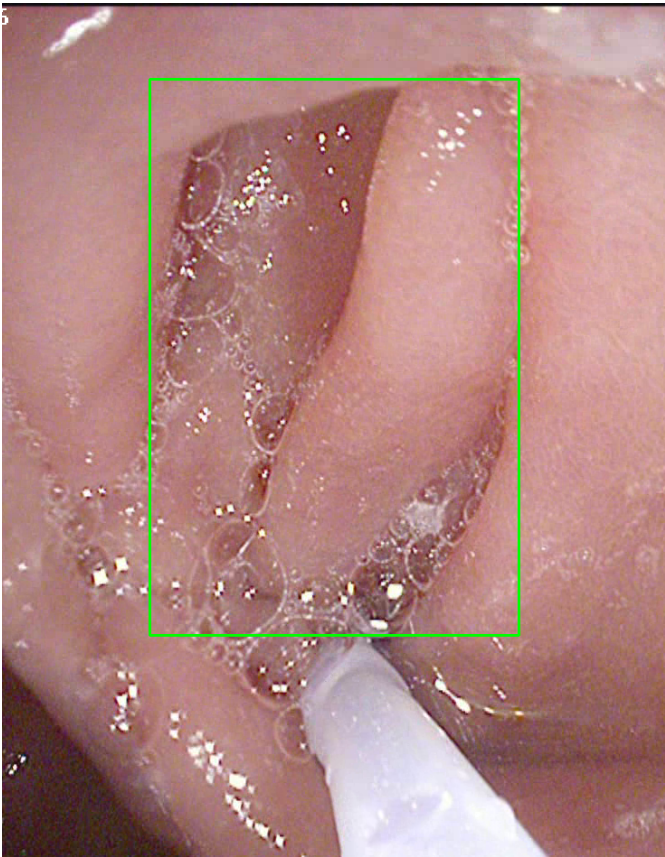
Feature	Value	Interpretation
Frame Index	417	Frame 417 of 3607
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 84.1%
Redness Score	0.148	Color-based vascularization
Relative Radius	16.9%	Polyp size as % of frame diagonal
Texture Score	0.046	Surface morphology
Vessel Visibility	0.426	Vascularization indicator
Risk Tier	MEDIUM_RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket

Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	67.7%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 200 frames)



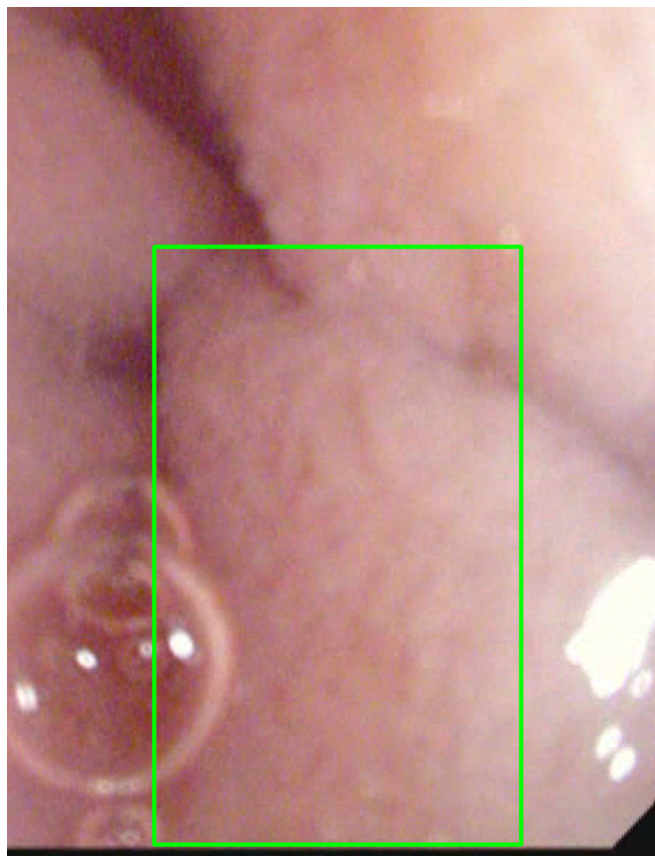
Feature	Value	Interpretation
Frame Index	2167	Frame 2167 of 3607
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	SERRATED_P OLYP	Type confidence: 75.5%

Redness Score	0.124	Color-based vascularization
Relative Radius	40.7%	Polyp size as % of frame diagonal
Texture Score	0.119	Surface morphology
Vessel Visibility	0.197	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	73.2%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 143 frames)



Feature	Value	Interpretation
Frame Index	417	Frame 417 of 3607
Detection Method	Detector Consensus (MedSAM2 / RT-DETR)	Detector Consensus (MedSAM2 / RT-DETR) (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 84.1%
Redness Score	0.148	Color-based vascularization
Relative Radius	16.9%	Polyp size as % of frame diagonal
Texture Score	0.046	Surface morphology
Vessel Visibility	0.426	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	63.4%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.